📒 Insight Summary

🚢 Titanic Dataset

- **Gender impact**: Women had significantly higher survival rates than men, making gender the strongest predictor of survival.
- **Passenger class**: First-class passengers were more likely to survive compared to those in second and third class, showing the influence of socioeconomic status.
- **Fare correlation**: Higher ticket fares were associated with better survival chances, reflecting access to safer cabins and lifeboats.
- **Age patterns**: Children had better odds of survival compared to adults, highlighting prioritization during evacuation.
- **Engineered features**: FamilySize and Title extraction added predictive power by capturing social ties and passenger status.

🎯 Key Takeaways

- Clean, structured data reveals **clear survival drivers** in Titanic and **pricing determinants** in housing.
- **Demographics and socioeconomic factors** shaped Titanic outcomes.
- **Structural features and neighborhood context** shaped housing prices.
- These insights provide a strong foundation for predictive modeling in later stages.

Week 3 Conclusion

In this week's assignment, we applied **statistics and probability** concepts to the Titanic dataset:

- **Descriptive statistics** provided a clear overview of passenger demographics, showing an average age of ~29 years and highly variable ticket fares.
- **Probability distributions** revealed that Age followed a near-normal distribution, while Fare was right-skewed with significant outliers.
- **Hypothesis testing** (Chi-square test) confirmed that survival rates differed significantly between genders, validating earlier exploratory insights.
- **Correlation analysis** highlighted relationships between features such as Fare, Pclass, and survival. Importantly, we emphasized that correlation does not imply causation — socioeconomic status and evacuation priority were the true underlying drivers.

Key Takeaway

This week reinforced the importance of **statistical thinking** in data science. By combining descriptive measures, probability distributions, hypothesis testing, and correlation analysis, we built a strong foundation for understanding data behavior and guiding predictive modeling decisions in future assignments.

Submission Note (Weeks 1–3)

Dear Instructor,

I would like to submit my work for Weeks 1–3. For Weeks 1–2, I completed the full data cleaning and exploratory analysis pipeline using the Titanic dataset. Unfortunately, I experienced persistent technical issues with Kaggle login, which prevented me from accessing the House Prices dataset in time. I attempted multiple fixes but was unable to resolve the problem before the deadline.

To ensure I stayed on track, I focused on the Titanic dataset and demonstrated all required steps: handling missing values, duplicates, univariate and bivariate analysis, visualizations, and insights. For Week 3, I extended this work into statistical analysis and modeling, applying descriptive statistics, probability distributions, hypothesis testing, correlation analysis, and predictive models such as Logistic Regression and Random Forest.

While this submission is later than intended due to the Kaggle issue, I have documented my methodology thoroughly and ensured that the deliverables (cleaned dataset, Jupyter Notebook, insight summary, and repository updates) are complete. Once Kaggle access is restored, I plan to apply the same pipeline to the House Prices dataset and update my repository accordingly.

Best regards,
Charity.